UniHive™

THE CONNECTED MULTI-CAMPUS MARKETPLACE & DISPATCH NETWORK

OFFICIAL MASTER PROPOSAL
Version: **2.4 Enterprise**
Engine: **Node.js • React 19 • Socket.io • PostgreSQL**
Direct Contact: **idokoonu0@gmail.com**

Transforming Higher-Education Commerce into a Secure, High-Speed Campus Economy

A comprehensive institutional partnership proposal for University Leadership, Collegiate Student Councils, and Strategic Enterprise Partners to deploy UniHive™ as the official digital infrastructure for trusted student-to-student transactions, structured hostel delivery logistics, and secure escrow settlement.

100%	55/55	3 Tiers	₦0.00
ESCROW PROTECTION	PASSING CORE TESTS	CAMPUS DELIVERY MODES	INSTITUTIONAL SETUP COST

1. Executive Summary & Strategic Vision

Across tertiary educational institutions in Nigeria and Sub-Saharan Africa, students conduct hundreds of millions of Naira in commerce each semester—buying textbooks, electronic accessories, groceries, fashion, and hostel provisions. Yet, **more than 94% of this trade currently occurs over unmonitored WhatsApp groups, Telegram channels, and unstructured social forums**.

This informal commerce architecture is severely broken, presenting recurring crises for both students and university administration:

- **Rampant Impersonation & Payment Scams:** Buyers send bank transfers to unvetted accounts, only for the "seller" to block their number.
- **Hostel Inaccessibility & Security Risks:** Strict university hostel regulations (especially male-into-female hall curfews and security checkpoints) leave deliveries stranded or expose halls to unauthorized entry.
- **Lack of Regulatory Governance:** University management possesses zero audit trail, resolution mechanism, or institutional visibility into high-volume economic activity occurring on its property.

> **The UniHive™ Solution:** A closed-loop, multi-campus digital ecosystem that pairs a unified e-commerce marketplace with **real-time WebSocket dispatch tracking**, **automated porter-lodge logistics**, and an **unbreakable dual-handshake financial escrow** powered by Paystack.

2. Complete Cross-Platform Suite Showcase

Unlike theoretical pitch proposals, **UniHive™ is already fully engineered, tested, and demonstrated end-to-end**. Below is the multi-device interface suite built in React 19, Tailwind CSS, and TypeScript:

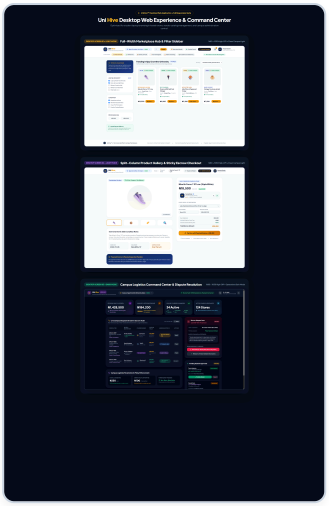


Figure 1.1: UniHive™ Desktop Suite — Integrated Marketplace Hub, Dynamic Multi-Item Checkout, and Executive Administrative Command Console.

3. The 4-Pillar Unified Campus Ecosystem

UniHive™ unifies four distinct campus personas into a single synchronized operational network:

Persona	Portal / Interface	Core Capabilities & Operational Flow
1. Student Buyer	Mobile PWA & Desktop Web Store	Hostel-filtered catalogue, real-time product search, multi-vendor cart, secure Paystack checkout, live dispatch GPS tracking, and delivery PIN generation.
2. Student Vendor	Vendor Studio Dashboard	Automated inventory stock control, instant live order push notifications, receipt printouts, pickup readiness marking, and automated escrow bank payouts.
3. Student Dispatcher (Runner)	Campus Courier Dispatch Portal	Open delivery claims pool, route-optimized multi-vendor pickups, porter lodge verification, and delivery completion confirmation.
4. University Administrator	Admin Command Console	Institutional oversight, KYC student identity verification, 2FA biometric security, live dispute resolution, and university commercial telemetry.

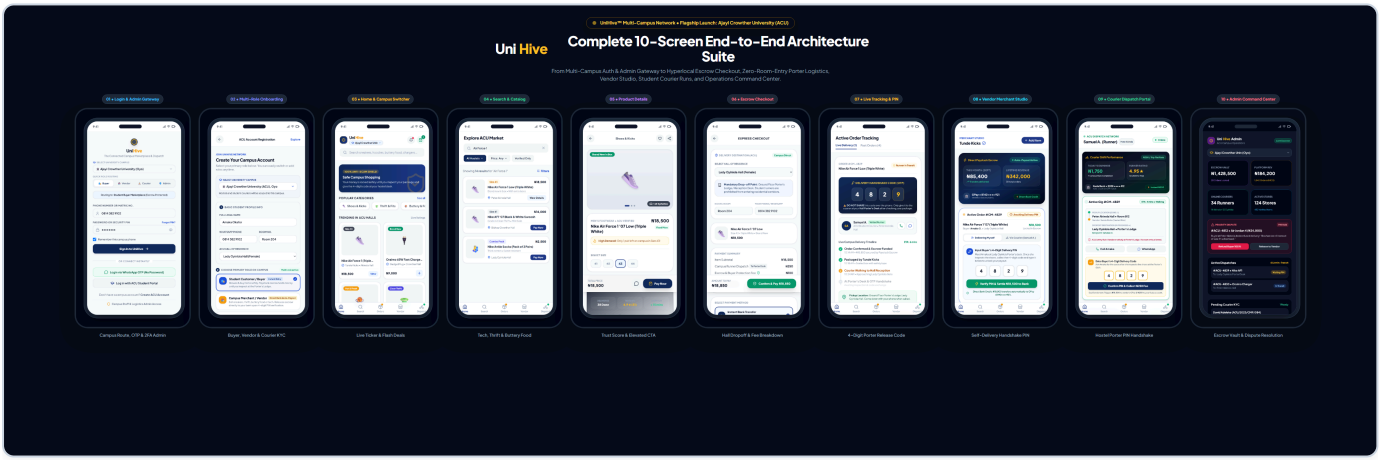


Figure 2.1: UniHive™ Mobile Web Suite — Complete screen flow from Discovery to Express Checkout, PIN Handshake, and Runner Dispatch.

4. Advanced Multi-Tier Delivery Architecture

Recognizing the diverse physical layouts and financial realities of Nigerian campuses, UniHive™ introduces **three tailored fulfillment modes**:

₦0
MODE 1: CAMPUS SELF-PICKUP
Direct exchange at verified campus landmarks (e.g., Chapel Quad, Faculty Hall). Zero delivery fees.

₦500
MODE 2: BATCH EVENING DROP
Orders consolidated and dispatched during peak hostel hours (6:00 PM – 9:30 PM) to hall porter desks.

₦1,500
MODE 3: EXPRESS RUNNER
Priority point-to-point delivery within 25 minutes by a dedicated student runner directly to the hostel lodge.

> **Multi-Vendor Universal Cart:** Students can add a textbook from a vendor in Hall A and snack supplies from a seller in Hall B in a single checkout session. The backend automatically segments sub-orders, generates discrete package identification tags, and routes individual collection tasks to dispatchers without duplicate payment fees.

5. Unbreakable Financial Escrow Architecture

To eliminate marketplace fraud completely, UniHive™ enforces an unalterable, automated **3-Way Escrow Settlement Cycle**:

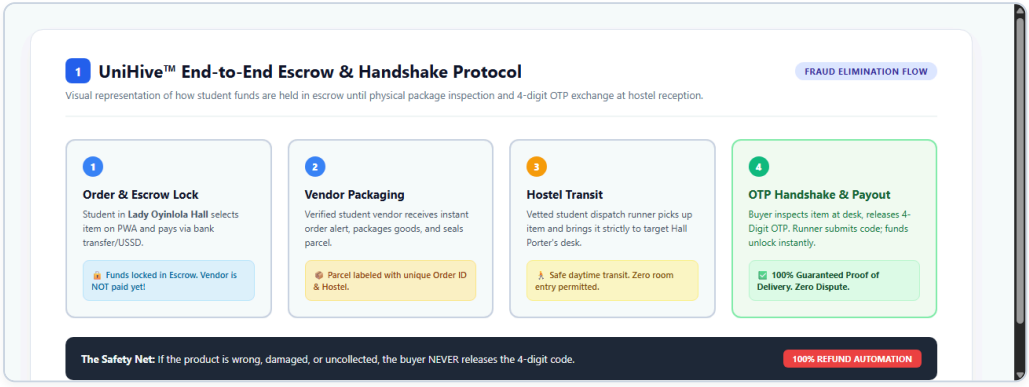


Figure 3.1: Automated Escrow Lifecycle — Cryptographic PIN Verification and Instant Paystack Multi-Split Settlement.

1. **Payment Isolation:** When a buyer checks out, funds are collected via Paystack and held in an isolated escrow vault. Neither the vendor nor the courier can access the money prematurely.
2. **Dual-Settlement PIN Handshake:** The buyer's application generates a unique, one-time 4-digit numeric token. The bcrypt hash of this token is stored in the database.
3. **Physical Inspection & Exchange:** Upon receiving the physical package at their hostel porter desk, the buyer inspects the goods. If satisfied, they hand the 4-digit code to the vendor or runner.
4. **Instant Micro-Payouts:** Entering the correct PIN into the Vendor/Courier app triggers instant atomic disbursement:
    - **Vendor:** 100% of product price credited instantly.
    - **Courier:** 100% of delivery fee credited instantly.
    - **UniHive:** ₦100 flat tech maintenance fee credited to platform reserve.

6. Hostel Regulations & Porter-Lodge Security Integration

Strict gender segregation and safety policies are institutional hallmarks of university residential life. UniHive™ respects and strengthens these protocols:

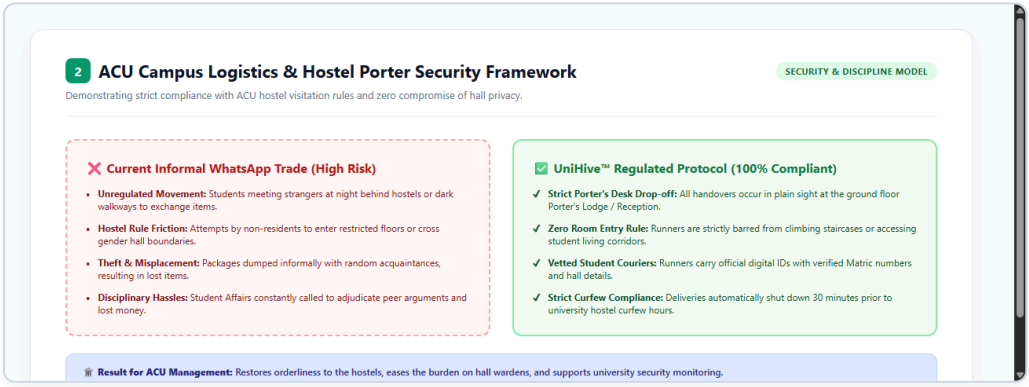


Figure 3.2: Gender-Segregated Hall Delivery Protocol & Porter Lodge Handshake System.

> **Zero Room-Entry Policy:** Couriers and external vendors are strictly prohibited from entering student rooms or cross-gender corridors. All handovers terminate at the **Ground-Floor Porter Reception Lodge**. Hall porters oversee package handoffs, ensuring safety, student peace of mind, and full compliance with student affairs policies.

7. Anti-Impersonation & Student Identity Verification

Security is the foundation of university adoption. UniHive™ eliminates the anonymity that plagues generic social networks through multi-layer verification:

- **Institutional Email Verification (`.edu.ng`):** Accounts require authentication via high-entropy 6-digit OTPs sent directly to valid institutional email addresses (e.g., `matric@acu.edu.ng`).
- **Matriculation Number Validation:** Student vendors and dispatchers must submit official matriculation records cross-referenced with university registries.
- **Hostel Residence Verification:** Profiles are pinned to specific university halls and room allocations, preventing outside infiltration.

> **Proven Technical Resilience — 55/55 Automated Test Suites Passing:**
> The UniHive™ backend undergoes rigorous automated quality assurance tests on every build. Core modules tested include: *Cart checkout calculations, Paystack webhook idempotent handling, PIN handshake rate-limiting (3 failed attempt freeze), stock decrement race conditions, WebSocket real-time broadcast integrity, and automated transactional email generation.*

8. Value Proposition for University Administration

Strategic Priority	Without UniHive™ (Current State)	With UniHive™ (Institutional Deployment)
Student Safety & Hall Security	Unchecked strangers roaming hostel perimeters for drop-offs.	100% vetted student runners, restricted to ground-floor porter lodges.
Fraud & Conflict Resolution	Dean of Student Affairs bogged down by fraud complaints.	Automated financial escrow; funds remain locked until buyer inspects items.
Student Entrepreneurship	Student businesses struggle with marketing and logistics.	Professional digital storefronts with analytics, inventory, and instant payouts.
On-Campus Employment	Limited part-time work opportunities for working students.	Flexible, lucrative student courier roles earning ₦500 – ₦1,500 per delivery trip.
University Governance & Data	Total blindness into commercial campus activity.	Dedicated University Admin Console tracking real-time campus commerce telemetry.

9. Phased Implementation Roadmap

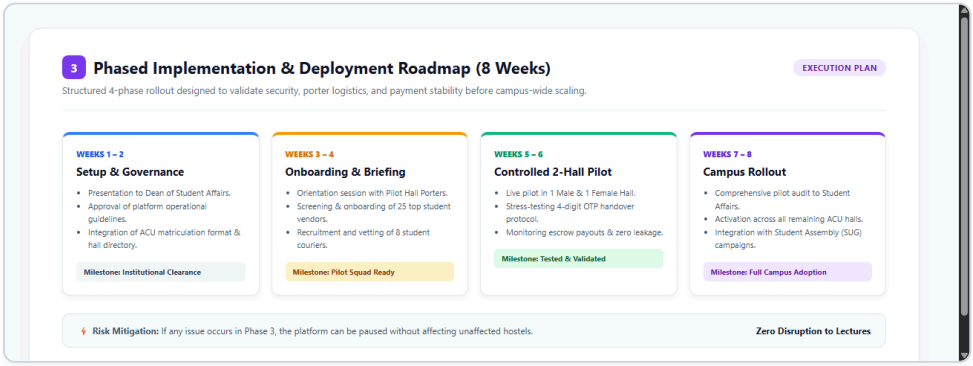


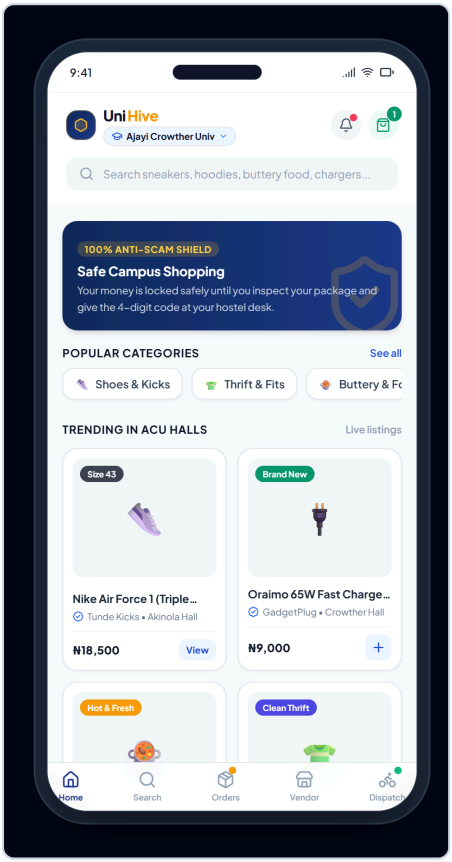
Figure 4.1: 4-Phase Rollout Roadmap from ACU Campus Pilot to National Multi-University Deployment.

Submitted by:
UniHive™ Development Team
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Official Inquiries: **idokoonu0@gmail.com**

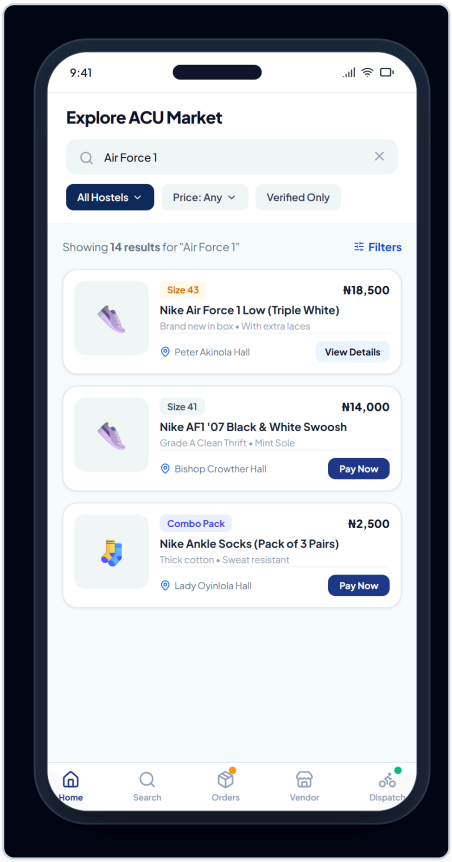
Institutional Readiness:
PRODUCTION READY (V2.4)
Platform: **Web • Android PWA • iOS PWA**
Deployment Timeline: **2 Weeks Post-Approval**

10. Complete UI/UX Screen Gallery & Operational Roles

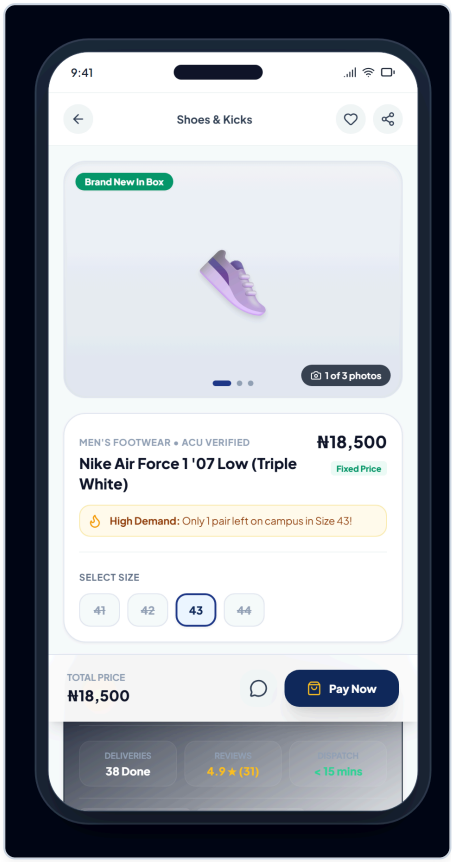
The following visual gallery documents the production user interface across each stage of the campus commerce lifecycle:



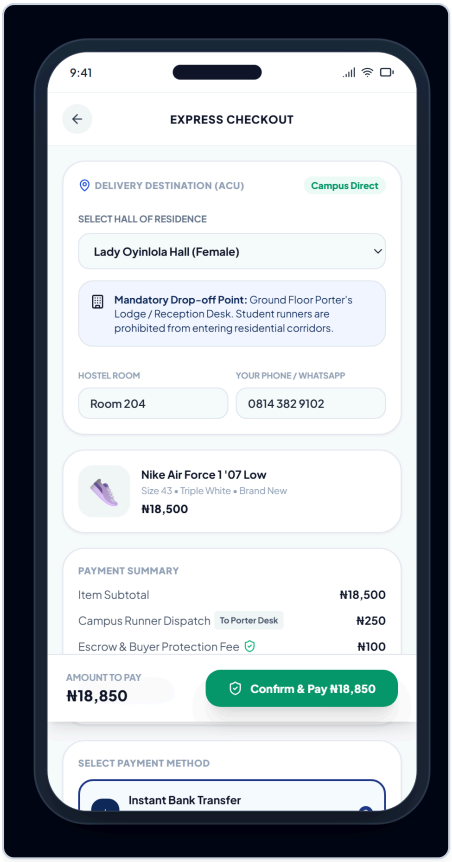
Screen 1: Discovery & Feed



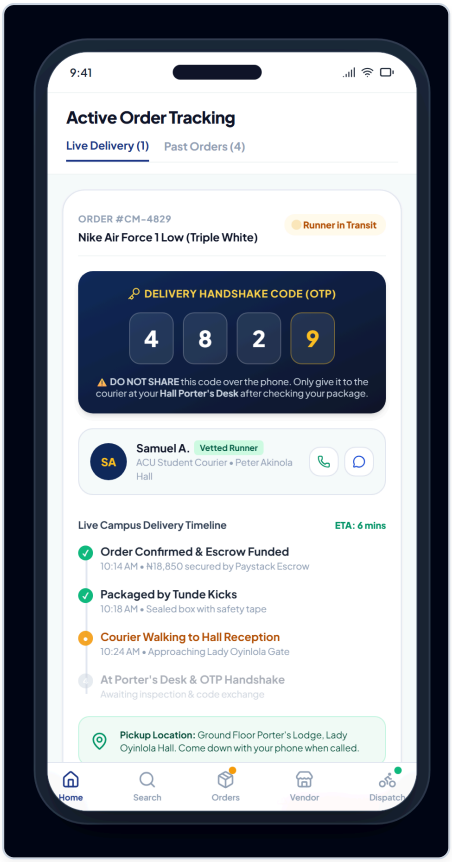
Screen 2: Category Search



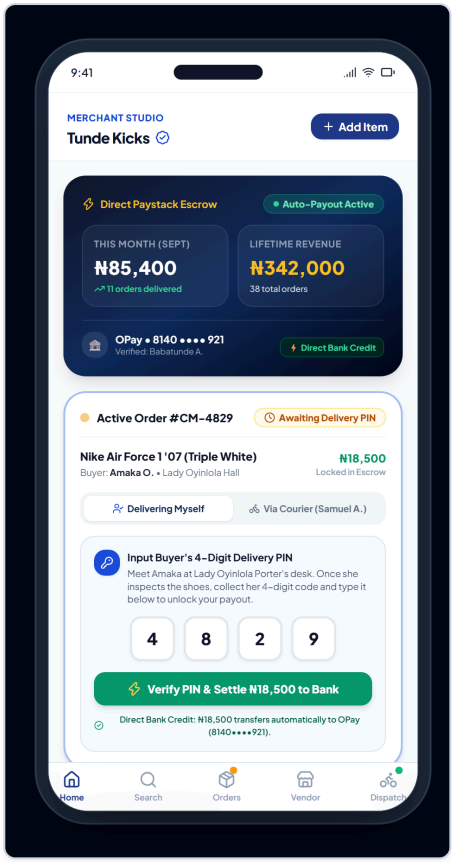
Screen 3: Product Details & Cart



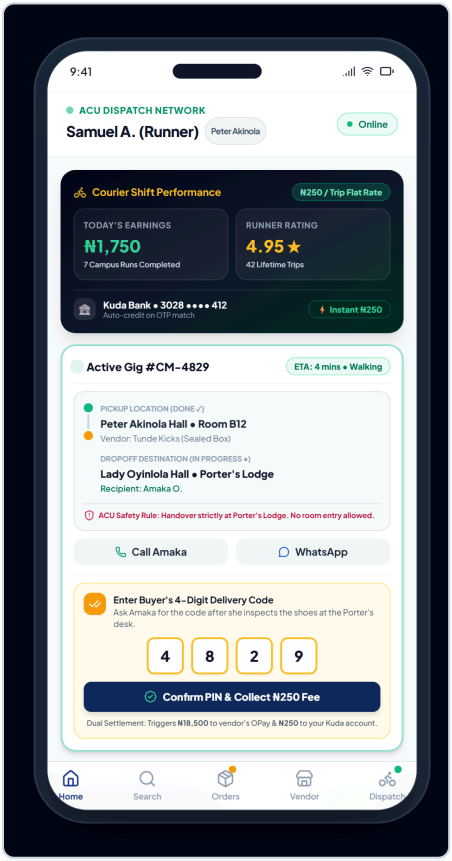
Screen 4: Paystack Checkout



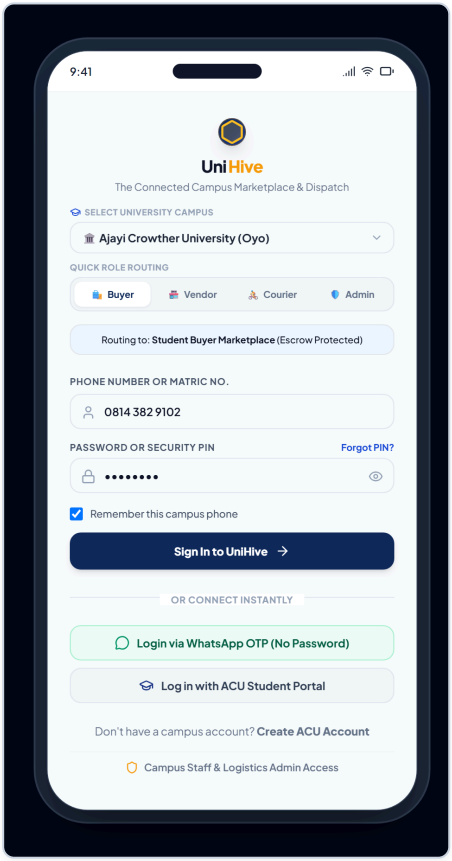
Screen 5: Live WebSocket GPS



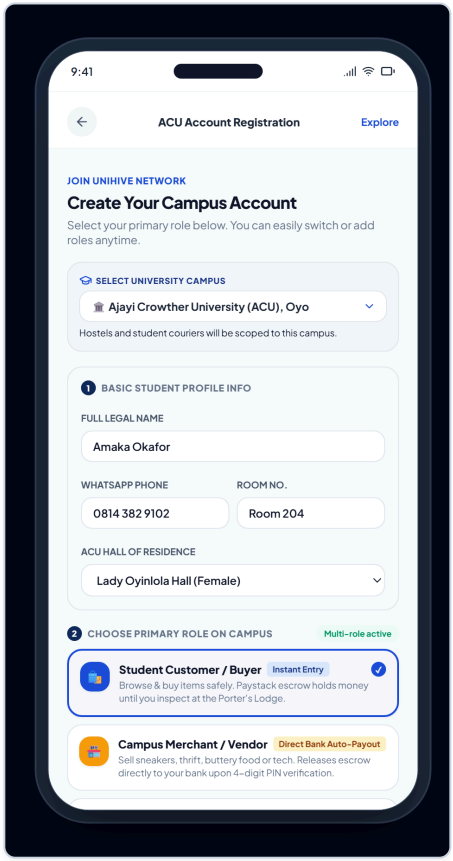
Screen 6: Vendor Studio & PIN Entry



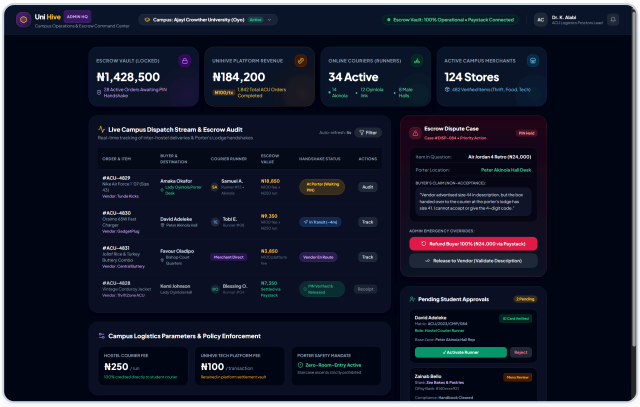
Screen 7: Runner Dispatch Portal



Screen 8: Biometric / PIN Login



Screen 9: Role Onboarding



Screen 10: Institutional Admin Dashboard — Live Campus Revenue, Dispute Radar, and Courier Governance.